

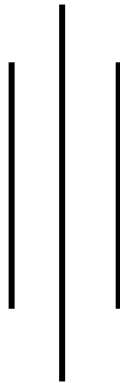
NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari Lalitpur



(Affiliated to Pokhara University)

Case Study on “IoT Based Theft Monitoring System”



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1. INTRODUCTION

IoT (Internet of Things) refers to the network of interconnected devices like sensors, actuators that communicate and exchange data over Internet in order to produce certain task with or without human intervention. These devices collect, process, and exchange data automatically, enabling real-time monitoring and decision-making. IoT has transformed many sectors like healthcare, agriculture, public safety and many more.

In today's world, theft remains a significant concern for individuals and businesses, leading to substantial financial losses and emotional distress. Traditional security systems often have limitations, such as fixed surveillance points and delayed response times. The advancement of IoT technology offers innovative solutions to address these challenges by enabling proactive and real-time theft monitoring.

The IoT Based Theft Monitoring System is an intelligent solution designed to detect and deter theft in various environments, such as homes, offices, warehouses, or vehicles. The system utilizes a combination of sensors, cameras, GPS, and IoT connectivity to identify suspicious activities. It can detect unauthorized entry, movement of valuable assets, or unusual environmental changes that might indicate a theft attempt.

This project combines embedded systems, cloud computing, and various IoT technologies to solve the real-world problem of theft. It can send real-time alerts and location data to authorities or designated personnel through cloud-based platforms. This project demonstrates how IoT can significantly improve security and reduce the incidence of theft, providing an extra layer of protection for valuable assets.

2. BACKGROUND STUDY

Theft incidents often occur due to a lack of continuous surveillance or delayed responses from traditional security measures. Existing security systems, such as alarms and fixed CCTV cameras, provide some level of deterrence but often fail to offer real-time, comprehensive monitoring and immediate intervention. For instance, traditional CCTV systems provide surveillance only at fixed locations and cannot actively track moving objects or suspicious individuals beyond their field of view.

Several problems might already exist. Some of them are:

- Lack of continuous surveillance in vulnerable areas.
- Difficulty in sharing real-time location of stolen items or intruders during emergencies.
- Dependence on manual reporting or delayed notification systems.
- Increasing incidents of theft, burglary, and unauthorized access.
- Delay in emergency response during critical situations.

Some of the current solutions that might be useful are:

- GPS-based live location tracking of assets.
- Autonomous monitoring of high-risk areas.
- Wi-Fi alerts to emergency contacts and security personnel.
- Cloud-based monitoring and data storage.
- Real-time video surveillance using cameras.

There already may be certain security measures for safety like CCTV surveillance, security guards, and alarm systems. Even though there are certain security measures they may have certain limitations. Some of them are: Limited coverage, Slow emergency responses, No autonomous monitoring and many other.

To control all those IoT technologies could be used. IoT could provide a better solution simply by integrating sensors, cloud computing, communication networks and certain intelligent monitoring. The IoT Theft Monitoring System could continuously send data to a central API so that continuous monitoring could be done with or without human intervention.

If any suspicious act is detected, it may simply alert certain security personnel. IoT basically helps reduce response time, enhances situational awareness, and creates a safer environment for protecting valuable assets. Certain reasons why IoT is needed for theft monitoring are:

- Sends instant alerts to owners and security personnel.
- Supports continuous surveillance using cameras and sensors.
- Stores and analyzes data through cloud platforms.
- Improves overall security and emergency management.

3. PROBLEM STATEMENT

Valuable assets are frequently vulnerable to theft due to a lack of continuous monitoring in isolated or unsupervised places. Security personnel cannot monitor every location continuously, and traditional surveillance systems are often limited to fixed areas.

An IoT-based Theft Monitoring System addresses this problem by providing continuous, autonomous surveillance. The system monitors designated areas, detects unauthorized access or movement, captures live video, determines the GPS location of assets, and sends instant notifications to security personnel or owners. The system enables faster emergency response, continuous surveillance, and improves situational awareness. With the growing need for better security, intelligent surveillance systems are becoming a vital solution. By enhancing public safety and reducing response time during emergencies, the system contributes to creating a safer and more secure environment for protecting assets.

The increasing number of theft cases has made many individuals and businesses feel insecure about their valuable possessions. This IoT-based Theft Monitoring System can contribute to creating a safer environment, preventing crimes, and improving the overall sense of security.

Our system can benefit many individuals:

- Homeowners
- Business owners
- Warehouse managers
- Vehicle owners
- Security agencies
- Police departments

4. SYSTEM ARCHITECTURE

The proposed IoT-based Theft Monitoring System follows a three-layer architecture: Device layer, Network layer, and Cloud layer. The system continuously monitors the predefined areas in real-time. Once the system detects anything suspicious, it sends alerts to designated security personnel or owners to prevent theft.

1. Device layer

The Device layer consists of hardware components like ESP32 microcontroller, Raspberry Pi, GPS module, Camera module, and various sensors (e.g., PIR motion sensor, ultrasonic sensor, vibration sensor, door/window sensors) to detect intrusions or unauthorized movement. All of these devices continuously monitor the surroundings and send the data through REST API and are notified to the security personnel. The system includes sensors and modules like:

- Camera Module for live video monitoring and image capture.
- PIR Motion Sensor/Vibration Sensor for detecting unauthorized movement or tampering.
- Ultrasonic sensors for obstacle detection (if mobile) or proximity sensing.
- GPS Module to track down the location of assets or the monitoring device.
- ESP32 microcontroller to collect data and process the information.

2. Network layer

The Network layer provides communication between the monitoring devices and the certain dashboard that is created in cloud services. The ESP32 is connected to the internet using Wi-Fi or cellular modules, enabling real-time data transmission. Several communication protocols that are used are:

- **MQTT** (Message Queuing Telemetry Transport): The ESP32 publishes sensor readings, GPS location, and emergency alerts to the MQTT broker. The mobile application, web dashboard, or cloud platform subscribes to these topics and instantly receives updates.
- **HTTP** (Hyper Text Transfer Protocol): Used to send sensor readings, GPS coordinates, and emergency information from the ESP32 or Raspberry Pi to a cloud server.
- **REST API**: Enables mobile applications or web dashboards to retrieve the system's status, GPS location, and alert information.
- **WebSocket**: Supports real-time communication for live monitoring and instant notification updates.

3. Cloud layer

The Cloud layer is responsible for storing, processing, analyzing and managing the data received from IoT devices. In the Theft Monitoring IoT System, the cloud receives data like GPS location, sensor readings, emergency alerts, and camera images from the ESP32 or Raspberry Pi. Platforms like Google Cloud, AWS EC2 can be used depending on project requirements.

- Google Cloud offers scalable storage, analytics, and machine learning services.
- AWS EC2 hosts web servers, databases, and monitoring applications for large-scale deployments.

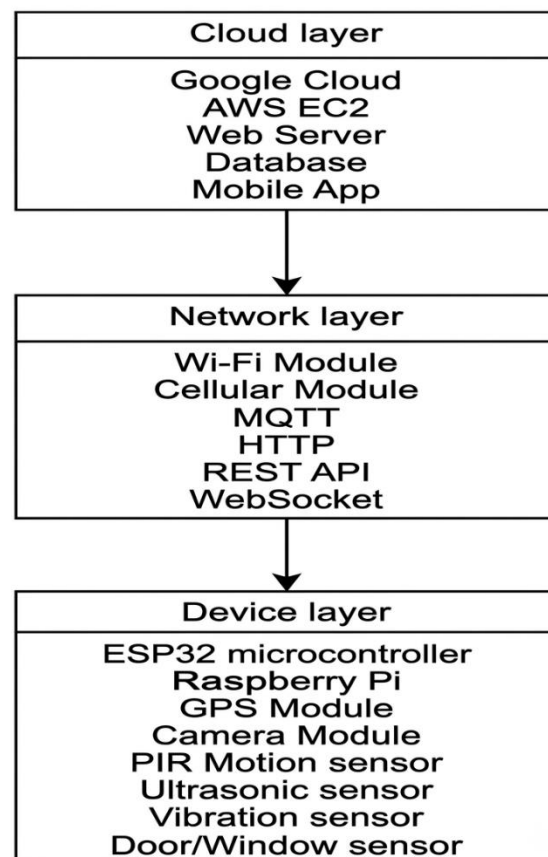
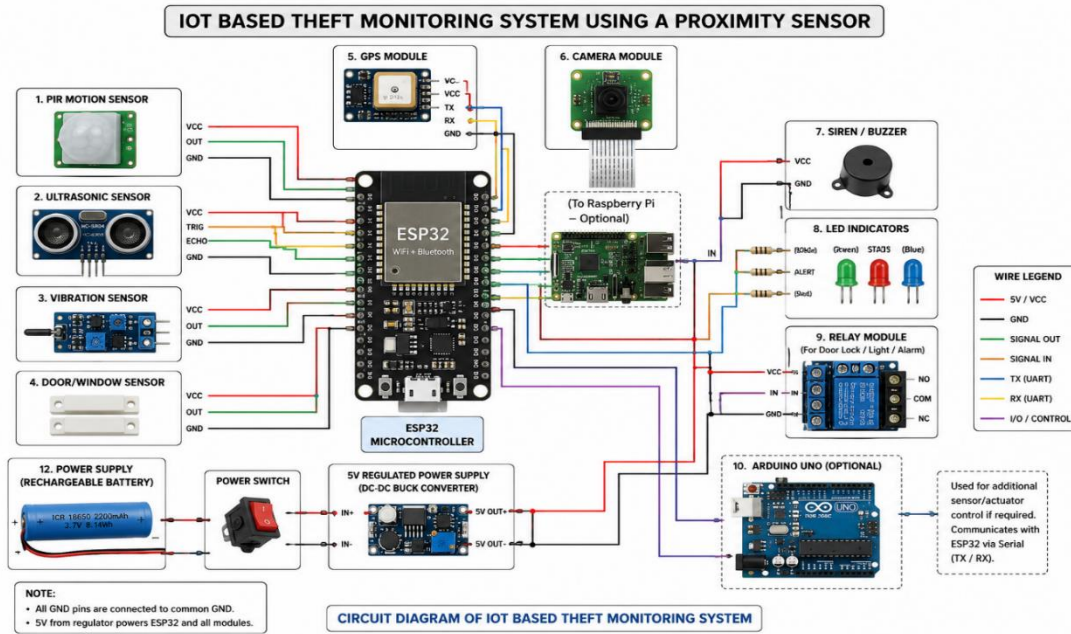


Fig 1: System Architecture

5. HARDWARE COMPONENTS

The proposed IoT-based Theft Monitoring System consists of the following hardware components:

- **ESP32 Microcontroller** — Acts as the main controller of the system. It processes sensor data and provides built-in Wi-Fi and Bluetooth for IoT communication.
- **Raspberry Pi (Optional)** — Used for advanced image processing, live video streaming, and AI-based object detection or facial recognition.
- **Arduino Uno (Optional)** — Controls sensors and actuators in systems where additional microcontroller support is required.
- **PIR Motion Sensor**— Detects human movement or unauthorized entry into a monitored area.
- **Ultrasonic Sensor** — Measures the distance to nearby objects, useful for detecting proximity or changes in an environment.
- **Vibration Sensor** — Detects physical tampering or movement of assets.
- **Door/Window Sensors** — Detects opening or closing of doors and windows.
- **GPS Module** — Determines the real-time geographical location of the monitoring device or asset and shares it during emergencies.
- **Camera Module** — Captures images or streams live video for remote monitoring and evidence collection.
- **Siren/Buzzer** — Provides an audible alarm to deter intruders.
- **LED Indicators** — Display the system status, power, or emergency conditions.
- **Rechargeable Battery/Power Supply** — Provides electrical power to the system and all connected components. These hardware components work together to detect theft attempts, communicate with cloud services, and provide real-time alerts, making the theft monitoring system an effective IoT-based security solution.



6. SOFTWARE COMPONENTS

The software components provide the programming, communication, monitoring and cloud services required for the theft monitoring system to function efficiently.

1. Programming language

For the IoT Based Theft Monitoring System, the primary programming language used is basically C/C++, written and uploaded using Arduino IDE. Also, Python could be used if Raspberry Pi alongside with camera and AI features are included.

1.1 C/C++: Used to program the ESP32, Arduino, sensors, and communication modules. This is the most commonly used language for microcontrollers.

1.2 Python: Used on Raspberry Pi for image processing, AI, camera control, GPS integration and cloud communication.

2. Frameworks

A framework is used to create web services (APIs), process data from sensors, and enable communication between the monitoring devices, cloud server and dashboard. The commonly used framework for IoT Based Theft Monitoring System is Flask.

2.1 Flask: A lightweight Python web framework which is used to build REST APIs that receive data from the monitoring devices and send it to the dashboard or cloud. Flask framework is suitable for real-time monitoring and emergency alert systems.

3. Database

A database simply is used for storing, organizing, managing of the data that are generated by the monitoring devices. It records data in the database for later use. Some of the databases are: SQLite, TinyDB, MySQL, PostgreSQL. But among all this SQLite is mostly recommended for this project.

3.1 **SQLite:** It is a lightweight, file-based relational database that stores sensor data, GPS coordinates, emergency alerts and event logs locally. Also SQLite is easier to integrate with Python and Flask.

4. Dashboard

It is a web-based interface allowing users to monitor and control the system in real-time which displays important information such as device's current GPS location, sensor status, emergency alerts, battery level and event history. The dashboard helps security personnel or owners quickly respond to emergencies and monitor the system remotely.

4.1 **HTML:** It is used to create the structure of the dashboard by organizing web pages by displaying buttons, tables, maps and status panels. HTML forms the basic layout for presenting monitoring information.

4.2 **JavaScript:** Adds interactivity and dynamic functionality to the dashboard by updating sensor values, GPS locations and emergency alerts in real time without reuploading the page.

4.3 **Chart.js:** Used to create graphical data visualizations that displays sensor readings, battery level, event statistics, and alert history using charts and graphs.

5. Cloud

A cloud is a platform that store data, process information, and provide remote access over the internet. The main cloud used to perform all those tasks are: AWS EC2, Microsoft Azure, Google Cloud Platform. But the main cloud used is AWS EC2.

5.1 **AWS (Amazon Web Services) EC2:** Provides virtual servers for hosting the IoT application, web server, APIs and database. It enables the monitoring devices to send sensor data and emergency alerts to the cloud in real time.

7. COMMUNICATION PROTOCOLS

Data should be exchanged between the monitoring devices, sensors, cloud server and dashboard which are possible using different communication protocols. Protocols ensure that the data are reliable and secure to be transmitted from a particular location to the particular dashboard. Some of the common communication protocols that could be used are:

7.1 MQTT (Message Queuing Telemetry Transport): Ideal for transmitting sensor readings, GPS coordinates and emergency alerts in real time.

7.2 HTTP (Hyper Text Transfer Protocol): Commonly used to send data to REST APIs that is developed using Flask framework. Follows a request-response model where the monitoring device or dashboard sends a request and the server returns a response.

7.3 WebSocket: Used to provide real-time update on the monitoring dashboard, such as the device's live GPS location, sensor readings and emergency alerts.

8. DATA FLOW ANALYSIS

Workflow description

When the system detects suspicious activity:

Firstly, PIR sensors detect human movement, vibration sensors detect tampering, and door/window sensors detect unauthorized entry. The ESP32 microcontroller then processes and collects all the sensor data. The Camera module and GPS module then provide live location and capture images or videos. The processed data is then transmitted over Wi-Fi (or cellular) where MQTT sends real-time IoT data while HTTP is used for REST API communication. The Flask framework then receives the data and forwards it to the cloud server. AWS EC2 then hosts the application and manages the incoming data. The SQLite database then stores sensor readings, GPS coordinates, emergency alerts, and event logs. The system's status, location, and alerts are then stored in the database. WebSockets can then push live updates to the dashboard without needing for the page to refresh. The security personnel/authorities/owners monitor the dashboard and immediately respond to the alerts.

9. SECURITY ANALYSIS

Possible threats

- **Unauthorized access:** Attackers may attempt to gain unauthorized access to the monitoring devices, cloud server or monitoring dashboard. Strong authentication and role-based access control help prevent unauthorized access.
- **Password attacks:** Weak or reused passwords can be easily hacked by hackers which may lead to access of the dashboard or cloud server by unauthorized user. Using strong passwords, multi-factor authentication may reduce the risk.
- **Device tampering/theft:** Since the monitoring devices may be deployed in various environments, they may be stolen or physically tampered with. An attacker could attempt to access stored data or damage the hardware.
- **Data tampering:** Attackers may intercept and alter sensor readings, GPS coordinates or emergency alerts during transmission. This could result in false alarms or incorrect monitoring information.
- **Network attacks:** The system communicates over Wi-Fi and the Internet, making it vulnerable to attacks like denial-of-service (DoS).

- **Security measures**

- **HTTPS:** HTTPS is used to secure the communication between the monitoring dashboard and the cloud server. It encrypts data transmitted over the Internet, preventing unauthorized users from intercepting sensitive information.
- **SSL/TLS:** SSL/TLS (Secure Sockets Layer/Transport Layer Security) encrypts data exchanged between the ESP32, cloud server, and web dashboard. It ensures the confidentiality and integrity of sensor data, GPS locations, and emergency alerts during transmission.
- **Authentication:** API keys provide secure access to the Flask REST API. Each authorized device is assigned a unique API key, ensuring that only trusted devices can send or retrieve data from the server.
- **Password Encryption:** User passwords are securely stored using strong hashing algorithms instead of plain text. This protects user credentials even if the database is compromised.
- **Firewall:** A firewall is configured on the AWS EC2 server to monitor and filter incoming and outgoing network traffic. It blocks unauthorized access while allowing only legitimate communication with the IoT system.

- **Secure Wi-Fi:** The monitoring device connects to the network using a secure Wi-Fi connection protected with WPA2/WPA3 encryption. This reduces the risk of unauthorized access and wireless attacks.
- **Access Control:** Role-based access control (RBAC) restricts system functions based on user roles. For example, administrators can configure the system, while security officers can monitor events and respond to alerts without modifying system settings.

10. MAPPING WITH COURSE LABS

- **Lab 01:** In lab 01 we simply hosted a web server on the cloud using AWS EC2 instance. And we were able to deploy a web page sample. Simply, by this lab we can host a web server along with a web dashboard, providing a cloud platform for receiving sensor data and allowing security authorities to monitor the system remotely.
- **Lab 02:** Here we developed and deployed a REST API to store data on a cloud environment. We connected to an AWS EC2 instance and installed certain required libraries and created FastAPI in main.py. A REST API was simply developed in order to receive sensor readings that were transmitted by ESP32 over Wi-Fi. The API stores the received information in the database, and this ensures that communication between the monitoring device and cloud server is reliable.
- **Lab 03:** In this lab we simply got to know about the various data visualization using certain API oriented architecture. And we created a dashboard which fetches the data from the API. A dashboard was created to visualize the sensor readings in real-time. The dashboard is made to display motion detection status, vibration alerts, GPS location, timestamps and system alerts, allowing operators to monitor the system continuously.
- **Lab 04:** We learned about different sensors, actuators, embedded system that could be used to carry out an IoT based project. The ESP32 was simply connected with PIR motion sensor, vibration sensor, and door/window sensors. ESP32 is responsible for collecting sensor readings, processing of the information and activating indicators like LED or a siren when suspicious movement or unauthorized access is detected.
- **Lab 05:** In Lab 05 we learnt about how to integrate ESP32 sensor data with cloud-based API and dashboard for visualization. The very final stage of this project is simply to connect every sensor in order for it to detect unauthorized access or theft attempts. The data that is obtained is later on transmitted via Wi-Fi to FAST API that is hosted on AWS EC2 and is later stored in the database and is displayed on the web dashboard. Security authorities/owners can then monitor the dashboard remotely and receive timely information for quick response.

11. CHALLENGES AND LIMITATIONS

Challenges

1. **Internet Connectivity:** The system depends on a stable Wi-Fi or internet connection to send sensor data and alerts to the cloud. If the internet connection is weak or unstable the data transmission may be disrupted or delayed, preventing the security authorities from receiving timely notification/alerts.
2. **Cybersecurity:** Protecting the system from unauthorized access, data tampering, and network attacks is a major challenge. Strong authentication, encryption (HTTPS/SSL), API keys, and secure access control are required to keep the system secure.
3. **Power Management:** ESP32, sensors used may consume battery power continuously and long hours of monitoring can quickly drain the battery of the device. If battery is depleted, the system may stop monitoring the area and fail to send emergency alerts.

Limitations

1. **Limited Detection Range:** The system can detect objects and movements within the range of its sensors. Obstacles like walls or large objects can block the sensors and create blind spots.
2. **Sensor Accuracy:** The sensors used may sometimes generate false alarms by detecting irrelevant movements or may produce incorrect measurements in certain environmental conditions.
3. **Regular Maintenance:** The system may require maintenance of the sensors, battery, and other hardware components. Dust, moisture, or damage can reduce the performance and accuracy of those components.

12. FUTURE IMPROVEMENTS

The Theft Monitoring System can be further enhanced by integrating technologies that improve its intelligence, reliability and efficiency. The system can also later on be backed up using 5G/6G, longer battery life through solar charging. Some of the other future improvements that could be done are:

- Integration of AI-based object recognition to identify specific valuable items or suspicious objects.
- Using of 5G/6G as a backup to Wi-Fi for uninterrupted connectivity.
- Implementing of machine learning to reduce false alarm alerts.
- Development of a mobile app to provide instant alerts and live monitoring for owners and security personnel.
- Integration of cloud analytics to analyze event data and identify high-risk patterns or locations.

13. PERSONAL REFLECTION

Personally, I feel that many valuable assets are not safe due to various factors. So basically, what is learnt is how an IoT system could be designed by any individual for theft protection. Also, I learned how an embedded system could be properly used alongside all the sensors and transmit the data to a cloud server using APIs and how we can simply display the information in real-time through a dashboard for security use. The skill that I gained helped me develop both technical and analytical skills. I learned how we can use sensors and dashboards in real-time and made it be useful for security purposes. The labs that we did also helped me a lot by making me able to launch an instance and how to use particular sensors alongside with different embedded systems.

This project has made me curious about how these tiny things (sensors, embedded systems) could be used to do such a big thing that are very useful for mankind. Also, I would like to explore more about advanced features like AI, GPS tracking, Machine learning. I also want to expand this particular project in order to support broader security and emergency response services.

14. CONCLUSION

We can simply conclude how this particular project could be used for theft monitoring. The system provides continuous surveillance and the cloud enables remote monitoring along with REST API enabling proper communication. Although the system has certain limitations, those could be easily addressed through advanced communication technologies, improved sensors, stronger mechanisms and many more. Future enhancements like AI-based threat detection, GPS tracking, machine learning and mobile application support can further improve the system's performance and reliability.

Overall, this case study demonstrates that an IoT-based Theft Monitoring System is a practical and scalable solution for enhancing security, enabling real-time monitoring and supporting faster emergency response. Wide range of area surveillance is done using cameras and various sensors. The IoT Based Theft Monitoring System offers a comprehensive solution to improve security by integrating these components, making environments safer and more secure for valuable assets.

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